

## 4.5 Duality for Subspaces and Quotients

A subspace  $U$  of  $V$  gives two new vector spaces:  $U$  itself, and the quotient  $V/U$  of Chapter 3. Each has a dual space, and so far  $U^*$  and  $(V/U)^*$  have nothing to do with  $V^*$ . This section connects them. The answer is a swap: the dual of the quotient is a **subspace** of  $V^*$ , namely the annihilator  $U^0$ , and the dual of the subspace is a **quotient** of  $V^*$ , namely  $V^*/U^0$ . Both identifications come from the First Isomorphism Theorem, and neither needs a basis. At the end we use them to describe hyperplanes, the subspaces cut out by a single equation, in any dimension.

### 4.5.1 Restricting functionals to a subspace

Let  $U$  be a subspace of  $V$ . A functional  $\varphi \in V^*$  measures every vector of  $V$ , so in particular it measures the vectors of  $U$ . Forgetting the rest gives the **restriction**  $\varphi|_U: U \rightarrow F$ ,  $\mathbf{u} \mapsto \varphi(\mathbf{u})$ , which is a functional on  $U$ . This defines the restriction map

$$\rho: V^* \rightarrow U^*, \quad \rho(\varphi) = \varphi|_U.$$

It is not a new kind of map. Let  $\iota: U \rightarrow V$ ,  $\iota(\mathbf{u}) = \mathbf{u}$ , be the inclusion. Its dual map (Definition 4.4.1) sends  $\varphi$  to  $\varphi \circ \iota$ , and  $(\varphi \circ \iota)(\mathbf{u}) = \varphi(\mathbf{u})$ . So  $\rho = \iota'$ , and in particular  $\rho$  is linear.

Two questions decide what  $\rho$  does. Which functionals restrict to zero? By Definition 4.2.1, exactly those in  $U^0$ . And does every functional on  $U$  come from one on  $V$ ? That is an extension problem, and it has a positive answer in every dimension.

#### Theorem 4.5.1. Dual of a Subspace

Let  $V$  be a vector space over  $F$ , of any dimension, and let  $U$  be a subspace of  $V$ . The restriction map  $\rho: V^* \rightarrow U^*$ ,  $\rho(\varphi) = \varphi|_U$ , is linear and surjective, and  $\ker \rho = U^0$ . Hence

$$V^*/U^0 \rightarrow U^*, \quad \varphi + U^0 \mapsto \varphi|_U$$

is a well-defined isomorphism, and  $U^* \cong V^*/U^0$ .

*Idea.* The kernel is the definition of  $U^0$  read aloud. The real content is surjectivity: given  $\psi$  on  $U$ , we must extend it to all of  $V$ . A basis tells us how. Extend a basis of  $U$  to a basis of  $V$ ; a linear map may be prescribed freely on a basis, so declare the extension to agree with  $\psi$  on the basis of  $U$  and to be 0 on the new basis vectors. Then the First Isomorphism Theorem turns “surjective with kernel  $U^0$ ” into the isomorphism.

*Proof.* We saw above that  $\rho = \iota'$  is linear. For  $\varphi \in V^*$ ,  $\rho(\varphi) = 0$  means  $\varphi(\mathbf{u}) = 0$  for every  $\mathbf{u} \in U$ , which is  $\varphi \in U^0$  by Definition 4.2.1. So  $\ker \rho = U^0$ .

*Surjective.* Let  $\psi \in U^*$ . By Theorem 1.8.3,  $U$  has a basis  $B_U$ . It is a linearly independent subset of  $V$ , so by Theorem 1.8.2 there is a basis  $B$  of  $V$  with  $B_U \subseteq B$ . By Theorem 3.1.6 there is a linear map  $\varphi: V \rightarrow F$  with

$$\varphi(\mathbf{b}) = \psi(\mathbf{b}) \quad \text{for } \mathbf{b} \in B_U, \quad \varphi(\mathbf{b}) = 0 \quad \text{for } \mathbf{b} \in B \setminus B_U.$$

Then  $\varphi|_U$  and  $\psi$  are linear maps  $U \rightarrow F$  that agree on the basis  $B_U$  of  $U$ , so  $\varphi|_U = \psi$  by the uniqueness part of Theorem 3.1.6. Thus  $\rho(\varphi) = \psi$ , and  $\rho$  is surjective. (If  $V$  is finite-dimensional, the finite versions Corollary 1.5.5, Theorem 1.6.5 and Theorem 3.1.5 suffice, and no choice principle is needed.)

By the First Isomorphism Theorem (Theorem 3.9.7),  $\varphi + \ker \rho \mapsto \rho(\varphi)$  is a well-defined isomorphism  $V^*/\ker \rho \rightarrow \text{im } \rho$ . Since  $\ker \rho = U^0$  and  $\text{im } \rho = U^*$ , this is the claimed isomorphism. ■

In words: **a functional on  $U$  is a functional on  $V$ , up to adding something that vanishes on  $U$ .** Two extensions of the same  $\psi$  differ by an element of  $U^0$ , and every element of  $U^0$  can be added.

When  $V$  is finite-dimensional, taking dimensions recovers the count of this chapter. By Corollary 4.1.4,  $\dim U^* = \dim U$  and  $\dim V^* = \dim V$ , and by Theorem 3.9.5,  $\dim(V^*/U^0) = \dim V - \dim U^0$ . So the isomorphism says  $\dim U = \dim V - \dim U^0$ , which is Theorem 4.2.2 again.

## 4.5.2 Functionals on a quotient

Now the other space. A functional  $\psi$  on  $V/U$  measures cosets. Composing with the quotient map  $\pi: V \rightarrow V/U$ ,  $\pi(\mathbf{v}) = \mathbf{v} + U$ , turns it into a measurement of vectors,  $\mathbf{v} \mapsto \psi(\mathbf{v} + U)$ . This functional on  $V$  cannot tell apart two vectors in the same coset, so in particular it is zero on  $U$ . The theorem says that nothing is lost and nothing else appears: the functionals on  $V/U$  are **exactly** the functionals on  $V$  that vanish on  $U$ .

### Theorem 4.5.2. Dual of a Quotient

Let  $V$  be a vector space over  $F$ , of any dimension, let  $U$  be a subspace, and let  $\pi: V \rightarrow V/U$  be the quotient map. Then the dual map

$$\pi': (V/U)^* \rightarrow V^*, \quad \pi'(\psi) = \psi \circ \pi,$$

is injective, and its image is  $U^0$ . Hence  $\pi'$  is an isomorphism from  $(V/U)^*$  onto  $U^0$ , and  $(V/U)^* \cong U^0$ .

*Idea.* Injectivity uses only that  $\pi$  is surjective: if  $\psi \circ \pi$  is zero, then  $\psi$  is zero on every coset. For the image, one inclusion is “ $\pi$  kills  $U$ ”. The other inclusion asks us to build a functional on  $V/U$  from a functional on  $V$  that vanishes on  $U$ , and that is exactly what the universal property of the quotient does.

*Proof.* The map  $\pi'$  is linear by Definition 4.4.1.

*Injective.* Suppose  $\pi'(\psi) = \psi \circ \pi = 0$ . Every element of  $V/U$  is  $\pi(\mathbf{v})$  for some  $\mathbf{v} \in V$ , because  $\pi$  is surjective (Theorem 3.9.4 (c)). Then  $\psi(\pi(\mathbf{v})) = 0$  for every  $\mathbf{v}$ , so  $\psi = 0$ . By Theorem 3.2.3,  $\pi'$  is injective.

$\text{im } \pi' \subseteq U^0$ . For  $\mathbf{u} \in U$ ,  $\pi(\mathbf{u}) = U$  is the zero vector of  $V/U$ , so  $(\psi \circ \pi)(\mathbf{u}) = \psi(\mathbf{0}) = 0$ .

Hence  $\psi \circ \pi \in U^0$ .

$U^0 \subseteq \text{im } \pi'$ . Let  $\varphi \in U^0$ . Then  $U \subseteq \ker \varphi$ , so by Theorem 3.9.10 (a) there is a linear map  $\psi: V/U \rightarrow F$  with  $\psi \circ \pi = \varphi$ . This  $\psi$  lies in  $(V/U)^*$ , and  $\pi'(\psi) = \varphi$ .

So  $\pi'$  is an injective linear map with image  $U^0$ ; regarded as a map onto  $U^0$  it is a linear bijection, hence an isomorphism. ■

Concretely, the inverse isomorphism  $U^0 \rightarrow (V/U)^*$  is  $\varphi \mapsto \bar{\varphi}$ , where  $\bar{\varphi}(\mathbf{v} + U) = \varphi(\mathbf{v})$ . The formula is well defined precisely because  $\varphi$  vanishes on  $U$ : if  $\mathbf{v} - \mathbf{v}' \in U$ , then  $\varphi(\mathbf{v}) - \varphi(\mathbf{v}') = \varphi(\mathbf{v} - \mathbf{v}') = 0$ .

The two theorems fit together as a mirror:

|             | subspace $U \subseteq V$                            | quotient $V/U$                                      |
|-------------|-----------------------------------------------------|-----------------------------------------------------|
| its dual is | a <b>quotient</b> of $V^*$ :<br>$U^* \cong V^*/U^0$ | a <b>subspace</b> of $V^*$ :<br>$(V/U)^* \cong U^0$ |
| the map     | $\iota'$ : restrict                                 | $\pi'$ : compose with $\pi$                         |
| the map is  | surjective                                          | injective                                           |

The last row is no accident:  $\iota$  is injective and  $\pi$  is surjective, and dualizing swaps the two properties.

**Alternatively**, in finite dimension both theorems follow from Theorem 4.4.3, which says  $\ker T' = (\text{im } T)^0$  and  $\text{im } T' = (\ker T)^0$ . For  $T = \pi$ :  $\text{im } \pi = V/U$ , whose annihilator is  $\{0\}$ , and  $\ker \pi = U$ , so  $\pi'$  is injective with image  $U^0$ . For  $T = \iota$ :  $\text{im } \iota = U$ , so  $\ker \iota' = U^0$ , and  $\ker \iota = \{0\}$ , whose annihilator is all of  $U^*$ , so  $\iota'$  is surjective. The proofs above avoid dimension counts, which is why they work in every dimension.

*Quick check.* Let  $V$  be a vector space of dimension 6 and  $U$  a subspace of dimension 2. Find  $\dim U^0$ ,  $\dim(V/U)^*$ ,  $\dim U^*$  and  $\dim(V^*/U^0)$ , and check that the two theorems are consistent with these numbers.

*Solution.* By Theorem 4.2.2,  $\dim U^0 = 6 - 2 = 4$ . By Theorem 3.9.5 and Corollary 4.1.4,  $\dim(V/U)^* = \dim(V/U) = 4$ , matching  $(V/U)^* \cong U^0$ . Next,  $\dim U^* = \dim U = 2$ , and  $\dim(V^*/U^0) = 6 - 4 = 2$ , matching  $U^* \cong V^*/U^0$ .

### 4.5.3 A line in $\mathbb{R}^3$ , in coordinates

The theorems name their isomorphisms without choosing anything. To see them act, we now choose bases and compute. Recall that a functional on  $\mathbb{R}^3$  is determined by its values on  $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$  (Theorem 4.1.1). So every  $\varphi \in (\mathbb{R}^3)^*$  has the form  $\varphi(x, y, z) = ax + by + cz$ , with  $(a, b, c) = (\varphi(\mathbf{e}_1), \varphi(\mathbf{e}_2), \varphi(\mathbf{e}_3))$ . We write such a functional simply as  $ax + by + cz$ .

**Example. Both isomorphisms for a line**

Let  $\mathbf{u} = (1, 2, -1)$  and  $U = \text{span}(\mathbf{u}) \subseteq \mathbb{R}^3$ .

- (a) Find a basis of  $U^0$ .
- (b) Find a basis of  $\mathbb{R}^3/U$ , its dual basis  $(\psi_1, \psi_2)$  of  $(\mathbb{R}^3/U)^*$ , and the images  $\pi'(\psi_1), \pi'(\psi_2)$ .
- (c) Describe the restriction map  $\rho: (\mathbb{R}^3)^* \rightarrow U^*$  and the isomorphism  $(\mathbb{R}^3)^*/U^0 \rightarrow U^*$ .

*Solution.*

- (a) By Theorem 4.2.3 (b),  $U^0 = \{\mathbf{u}\}^0$ , so  $ax + by + cz \in U^0$  exactly when  $a + 2b - c = 0$ , that is,  $c = a + 2b$ . Taking  $(a, b) = (1, 0)$  and  $(0, 1)$  gives

$$\varphi_1 = x + z, \quad \varphi_2 = y + 2z.$$

They are independent (look at the coefficients of  $x$  and  $y$ ), and  $\dim U^0 = 3 - 1 = 2$  by Theorem 4.2.2, so  $(\varphi_1, \varphi_2)$  is a basis of  $U^0$  (Theorem 1.6.4).

- (b) The list  $(\mathbf{u}, \mathbf{e}_1, \mathbf{e}_2)$  is independent: in  $a\mathbf{u} + b\mathbf{e}_1 + c\mathbf{e}_2 = (a + b, 2a + c, -a) = \mathbf{0}$ , the third entry gives  $a = 0$ , and then  $b = c = 0$ . It has length 3, so it is a basis of  $\mathbb{R}^3$ , and by Theorem 3.9.5  $(\mathbf{e}_1 + U, \mathbf{e}_2 + U)$  is a basis of  $\mathbb{R}^3/U$ . Its dual basis consists of the  $\psi_i$  with  $\psi_i(\mathbf{e}_j + U) = \delta_{ij}$  (Theorem 4.1.3).

The functional  $\pi'(\psi_1) = \psi_1 \circ \pi$  on  $\mathbb{R}^3$  takes the values  $\psi_1(\mathbf{e}_1 + U) = 1$  at  $\mathbf{e}_1$ ,  $\psi_1(\mathbf{e}_2 + U) = 0$  at  $\mathbf{e}_2$ , and  $\psi_1(U) = 0$  at  $\mathbf{u}$ . Writing it as  $ax + by + cz$ , this says  $a = 1$ ,  $b = 0$  and  $a + 2b - c = 0$ , so  $c = 1$ . Similarly  $\pi'(\psi_2)$  has  $a = 0$ ,  $b = 1$ ,  $c = 2$ . Hence

$$\pi'(\psi_1) = x + z = \varphi_1, \quad \pi'(\psi_2) = y + 2z = \varphi_2.$$

So  $\pi'$  carries the dual basis of  $(\mathbb{R}^3/U)^*$  onto the basis of  $U^0$  found in (a), as Theorem 4.5.2 predicts. Read backwards:  $\psi_1((x, y, z) + U) = x + z$  and  $\psi_2((x, y, z) + U) = y + 2z$ . These are well defined, since adding  $t\mathbf{u} = (t, 2t, -t)$  changes  $x + z$  by  $t - t = 0$  and  $y + 2z$  by  $2t - 2t = 0$ .

- (c) The space  $U^*$  is one-dimensional, with basis the functional  $\xi$  determined by  $\xi(\mathbf{u}) = 1$  (the dual basis of  $(\mathbf{u})$ ). A functional  $\varphi = ax + by + cz$  restricts to  $\varphi|_U = \varphi(\mathbf{u})\xi = (a + 2b - c)\xi$ , since both sides take the value  $a + 2b - c$  at  $\mathbf{u}$ . So  $\rho$  is “evaluate at  $\mathbf{u}$ ”. Its kernel is  $\{a + 2b - c = 0\} = U^0$ , and it is surjective, since  $\rho(x) = \xi$ . The isomorphism of Theorem 4.5.1 is

$$(ax + by + cz) + U^0 \mapsto (a + 2b - c)\xi.$$

For instance  $x + U^0 = -z + U^0$ , because  $x - (-z) = \varphi_1 \in U^0$ ; both restrict to  $\xi$ .

## 4.5.4 Why “natural” matters here

In the example, bases entered only when we wanted numbers. The maps  $\rho$  and  $\pi'$  were defined before any basis was chosen: restrict a functional, or compose it with  $\pi$ . That is what makes the two isomorphisms **natural**, in the sense of the discussion of the double dual. They are part of the structure of  $V$  and  $U$ , not of a coordinate system.

There is a tempting shortcut that is **not** natural. Since  $U^*$  is a quotient of  $V^*$ , one might try to find a copy of  $U^*$  **inside**  $V^*$ , as a subspace. This can be done, but only by making a choice. Pick

a complement  $W$  of  $U$ , so  $V = U \oplus W$ . A functional that vanishes on  $W$  is determined by its values on  $U$ , so the restriction  $\rho$  maps  $W^0$  isomorphically onto  $U^*$  ([Exercise \(C2: Complements and a choice of copy of  \$U^\*\$ \)](#)). But different complements give different subspaces  $W^0$ .

*Warning.*  $U^*$  is not a subspace of  $V^*$ . Its elements are functionals on  $U$ , which cannot be evaluated at vectors outside  $U$ . To place  $U^*$  inside  $V^*$  one must extend each functional, and extensions are not unique. In the example above, the complement  $W_1 = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$  of  $U$  gives  $W_1^0 = \text{span}(z)$ , while  $W_2 = \text{span}(\mathbf{e}_1, \mathbf{e}_3)$  gives  $W_2^0 = \text{span}(y)$ . These are different lines in  $(\mathbb{R}^3)^*$ , and both restrict isomorphically onto  $U^*$ :  $z|_U = -\xi$  and  $y|_U = 2\xi$ . The choice-free statement is  $U^* \cong V^*/U^0$ , a quotient.

The same caution applies on the other side, in reverse.  $(V/U)^*$  is naturally a subspace of  $V^*$ , namely  $U^0$ , but  $V/U$  itself is not a subspace of  $V$ . Duality turns the quotient, which has no natural home inside  $V$ , into a subspace, which does.

## 4.5.5 Codimension and hyperplanes

The quotient measures how far a subspace is from filling the whole space. In  $\mathbb{R}^3$  a plane through the origin misses “one direction” and a line misses “two directions”. For finite-dimensional  $V$  the number of missing directions is  $\dim V - \dim U$ . But this formula is useless in  $F[x]$ , where  $\dim V$  and  $\dim U$  may both be infinite, although  $U = \{p : p(0) = 0\}$  misses only one direction: every polynomial is a constant plus an element of  $U$ . The quotient gives the right number in every case.

*The codimension of  $U$  counts the directions in  $V$  that  $U$  is missing.*

### Definition 4.5.1. Codimension

Let  $U$  be a subspace of a vector space  $V$  over  $F$ . If  $V/U$  is finite-dimensional, the **codimension** of  $U$  in  $V$  is

$$\text{codim } U := \dim(V/U).$$

If  $V/U$  is infinite-dimensional, we say that  $U$  has **infinite codimension**.

In words: we do not compare the sizes of  $U$  and  $V$ . We collapse  $U$  to zero and measure what is left. The codimension depends on the ambient space, so when that is unclear we write “codimension of  $U$  in  $V$ ”.

### Definition 4.5.2. Hyperplane

A **hyperplane** in  $V$  is a subspace  $H$  of  $V$  of codimension **exactly** 1, that is, with  $\dim(V/H) = 1$ .

### Examples.

- **Finite dimension.** If  $V$  is finite-dimensional, then  $\text{codim } U = \dim V - \dim U$  by Theorem 3.9.5. The hyperplanes of  $F^n$  are its subspaces of dimension  $n - 1$ : the lines through  $\mathbf{0}$  in  $F^2$  and the planes through  $\mathbf{0}$  in  $F^3$ .
- **Trace.** In  $M_n(F)$  with  $n \geq 1$ , the trace-zero matrices are the kernel of  $\text{tr}: M_n(F) \rightarrow F$ . This map is linear and onto, since  $\text{tr}(c\mathbf{E}_{11}) = c$ , so  $M_n(F)/\ker \text{tr} \cong F$  by Theorem 3.9.7,

and the trace-zero matrices form a hyperplane.

- **Polynomials.** In  $F[x]$ , let  $U = \{p : p(0) = 0\}$ . Evaluation at 0 is a linear map  $F[x] \rightarrow F$  with kernel  $U$ , and it is surjective because the constant  $c$  maps to  $c$ . By Theorem 3.9.7,  $F[x]/U \cong F$ , so  $U$  is a hyperplane in the infinite-dimensional space  $F[x]$ . In the same way  $\{p : p(0) = p(1) = 0\}$  has codimension 2 ([Exercise \(B3: Dimension computations\)](#)).
- **Degenerate cases.**  $\text{codim } V = 0$ , since  $V/V$  is the zero space, and  $\text{codim}\{\mathbf{0}\} = \dim V$  when  $V$  is finite-dimensional. In particular  $V$  itself is **not** a hyperplane, and the only hyperplane of a one-dimensional space is  $\{\mathbf{0}\}$ . The zero space  $\{\mathbf{0}\}$  has no hyperplanes at all.

**Non-example by minimal change.** In  $\mathbb{R}^3$ , the plane  $\{x + y + z = 0\}$  is a hyperplane. Replace it by the line  $\{x + y + z = 0, x - y = 0\} = \text{span}((1, 1, -2))$ . It is still a subspace, but its codimension is  $3 - 1 = 2$ , so the clause “codimension **exactly** 1” fails. Replace it instead by  $\{x + y + z = 1\}$ : now it is not a subspace at all, so it has no codimension. It is a coset of a hyperplane, an object we return to in the next section.

**Why this definition.** Codimension is the dual notion of dimension, and the duality theorems make this precise. If  $U$  has finite codimension, then by [Theorem 4.5.2](#), Corollary 4.1.4 and Theorem 3.5.4,

$$\text{codim } U = \dim(V/U) = \dim(V/U)^* = \dim U^0,$$

and this holds even when  $V$  is infinite-dimensional. So **dimension counts vectors, and codimension counts independent equations**. For finite-dimensional  $V$  this is Theorem 4.2.2 again:  $\dim U^0 = \dim V - \dim U$ .

*Warning.* In infinite dimension a hyperplane need not be “smaller” than the whole space. In  $F[x]$ , the hyperplane  $U = \{p : p(0) = 0\} = \{xq : q \in F[x]\}$  is isomorphic to  $F[x]$  itself, via  $q \mapsto xq$ : this map is linear, injective (if  $xq = 0$  then  $q = 0$ ) and onto  $U$ . So you cannot detect codimension by comparing  $U$  with  $V$  up to isomorphism, and the formula  $\text{codim } U = \dim V - \dim U$  must never be used outside finite dimension. Always compute  $\dim(V/U)$ , or count equations.

Hyperplanes are exactly the subspaces cut out by **one** non-trivial linear equation. In  $F^n$  this is familiar:  $a_1x_1 + \cdots + a_nx_n = 0$  with the  $a_i$  not all zero defines a subspace of dimension  $n - 1$ . The quotient makes the statement true in every dimension.

### Theorem 4.5.3. Hyperplanes Are Kernels of Functionals

Let  $V$  be a vector space over  $F$ , of any dimension, and let  $H$  be a subspace of  $V$ . Then  $H$  is a hyperplane if and only if  $H = \ker \varphi$  for some **non-zero**  $\varphi \in V^*$ .

*Idea.* A non-zero functional maps onto the one-dimensional space  $F$ , so the First Isomorphism Theorem makes  $V/\ker \varphi$  one-dimensional. Conversely, if  $V/H$  is one-dimensional, it “is”  $F$  after choosing an isomorphism, and composing that isomorphism with  $\pi$  gives a functional whose kernel is  $H$ .

*Proof.* ( $\Leftarrow$ ) Suppose  $\varphi \in V^*$  is non-zero. Then  $\text{im } \varphi$  is a subspace of  $F$  (Theorem 3.2.2) containing a non-zero scalar, so  $\dim \text{im } \varphi = 1 = \dim F$ , and  $\text{im } \varphi = F$  by Theorem 1.6.8. By Theorem 3.9.7,  $V/\ker \varphi \cong F$ , so  $\dim(V/\ker \varphi) = 1$ , and  $\ker \varphi$  is a hyperplane.

( $\Rightarrow$ ) Suppose  $\dim(V/H) = 1$ . By Theorem 3.5.4 there is an isomorphism  $\theta: V/H \rightarrow F$ . Let  $\varphi = \theta \circ \pi$ , where  $\pi: V \rightarrow V/H$  is the quotient map. It is linear as a composition of linear maps (Theorem 3.4.2), so  $\varphi \in V^*$ . For  $\mathbf{v} \in V$ ,  $\varphi(\mathbf{v}) = 0$  means  $\theta(\mathbf{v} + H) = 0$ , which holds exactly when  $\mathbf{v} + H$  is the zero coset, since  $\theta$  is injective; and  $\mathbf{v} + H = H$  means  $\mathbf{v} \in H$  by Lemma 3.9.3 (b). So  $\ker \varphi = H$ . Finally  $\varphi \neq 0$ : since  $V/H \neq \{\mathbf{0}\}$ , we have  $H \neq V$ , so some  $\mathbf{v} \notin H = \ker \varphi$ . ■

The functional is not unique, but almost: two functionals with the same kernel are scalar multiples of each other (Exercise (C1: Functionals with the same kernel); [Exercise \(B2: Functionals with the same kernel\)](#) gives a second proof). So hyperplanes of  $V$  correspond to non-zero functionals **up to a non-zero scalar**, just as the plane  $x + y + z = 0$  is also  $2x + 2y + 2z = 0$ .

A hyperplane together with any one vector outside it fills the space.

*Remark.* Let  $H = \ker \varphi$  be a hyperplane and  $\mathbf{v}_0 \notin H$ . Then  $V = H \oplus \text{span}(\mathbf{v}_0)$ . Indeed  $\varphi(\mathbf{v}_0) \neq 0$ , and every  $\mathbf{v} \in V$  splits as

$$\mathbf{v} = \left( \mathbf{v} - \frac{\varphi(\mathbf{v})}{\varphi(\mathbf{v}_0)} \mathbf{v}_0 \right) + \frac{\varphi(\mathbf{v})}{\varphi(\mathbf{v}_0)} \mathbf{v}_0,$$

where the first bracket lies in  $\ker \varphi$  because  $\varphi$  of it is  $\varphi(\mathbf{v}) - \varphi(\mathbf{v}) = 0$ . And  $H \cap \text{span}(\mathbf{v}_0) = \{\mathbf{0}\}$ , since  $\varphi(c\mathbf{v}_0) = c\varphi(\mathbf{v}_0) = 0$  forces  $c = 0$ .

*Quick check.* Decide which of the following are hyperplanes, and for each hyperplane give a non-zero functional with that kernel.

- (a) The symmetric matrices in  $M_2(F)$ .
- (b) The symmetric matrices in  $M_3(F)$ .
- (c)  $\{p \in F[x] : p(1) = 0\}$  in  $F[x]$ .

*Solution.*

- (a) Yes.  $\mathbf{A}$  is symmetric exactly when  $a_{12} - a_{21} = 0$ , so the symmetric matrices are the kernel of  $\mathbf{A} \mapsto a_{12} - a_{21}$ , which is linear and non-zero (it takes the value 1 at  $\mathbf{E}_{12}$ ). By [Theorem 4.5.3](#) this is a hyperplane.
- (b) No. The symmetric  $3 \times 3$  matrices form a subspace of dimension 6 (choose the entries on and above the diagonal freely), so the codimension is  $9 - 6 = 3 \neq 1$ .
- (c) Yes. It is the kernel of  $p \mapsto p(1)$ , which is linear (Theorem 0.8.5) and non-zero (it sends the constant 1 to 1).

## 4.5.6 Exercises

### 4.5.6.1 A. Check your understanding

**Exercise. A1**

- (a) State the theorem identifying  $(V/U)^*$  with a subspace of  $V^*$ , including the map.
- (b) Define the codimension of a subspace, and define a hyperplane.
- (c) True or false:  $U^*$  is a subspace of  $V^*$ . Justify your answer.
- (d) Let  $U$  be a subspace of dimension 2 in  $F^5$ . Find  $\dim(F^5/U)^*$  and  $\text{codim } U$ .
- (e) Is  $\{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : x_1 = x_2 = 0\}$  a hyperplane in  $\mathbb{R}^4$ ? Justify your answer.
- (f) True or false: every non-zero linear functional  $\varphi: V \rightarrow F$  is surjective. Justify your answer.

*Solution.*

- (a) For a subspace  $U$  of  $V$  with quotient map  $\pi$ , the dual map  $\pi': (V/U)^* \rightarrow V^*$ ,  $\psi \mapsto \psi \circ \pi$ , is injective with image  $U^0$ ; so  $(V/U)^* \cong U^0$  (Theorem 4.5.2).
- (b) If  $V/U$  is finite-dimensional,  $\text{codim } U = \dim(V/U)$ ; otherwise  $U$  has infinite codimension. A hyperplane is a subspace of codimension exactly 1.
- (c) False. Elements of  $U^*$  are defined only on  $U$ , so they are not elements of  $V^*$ . What is true is  $U^* \cong V^*/U^0$ , a quotient (Theorem 4.5.1).
- (d)  $\dim(F^5/U)^* = \dim(F^5/U) = 5 - 2 = 3$ , by Corollary 4.1.4 and Theorem 3.9.5. So  $\text{codim } U = 3$ .
- (e) No. The subspace is  $\text{span}(\mathbf{e}_3, \mathbf{e}_4)$ , of dimension 2, so its codimension is  $4 - 2 = 2$ .
- (f) True, in any dimension. This is Proposition 4.1.2:  $\text{im } \varphi$  is a subspace of the one-dimensional space  $F$  containing a non-zero element, so it is  $F$ .

**4.5.6.2 B. Practice****Exercise. B1: The dual of a quotient of  $\mathbb{R}^4$** 

Let  $U = \text{span}((1, -1, 0, 2), (0, 1, 1, -1)) \subseteq \mathbb{R}^4$ , and write functionals on  $\mathbb{R}^4$  as  $a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4$ .

- (a) Find a basis  $(\varphi_1, \varphi_2)$  of  $U^0$ .
- (b) Let  $\psi_1, \psi_2 \in (\mathbb{R}^4/U)^*$  be the functionals with  $\pi'(\psi_i) = \varphi_i$ . Compute  $\psi_1$  and  $\psi_2$  at the coset  $(2, 0, 1, 3) + U$ .
- (c) Find a basis of  $\mathbb{R}^4/U$  whose dual basis is  $(\psi_1, \psi_2)$ .

*Solution.*

- (a) A functional lies in  $U^0$  exactly when it vanishes at both spanning vectors (Theorem 4.2.3 (b)), that is,

$$a_1 - a_2 + 2a_4 = 0, \quad a_2 + a_3 - a_4 = 0.$$

The second gives  $a_3 = a_4 - a_2$ , and the first gives  $a_1 = a_2 - 2a_4$ . Taking  $(a_2, a_4) = (1, 0)$  and  $(-1, -1)$  gives  $(1, 1, -1, 0)$  and  $(1, -1, 0, -1)$ , that is,

$$\varphi_1 = x_1 + x_2 - x_3, \quad \varphi_2 = x_1 - x_2 - x_4.$$



Check:  $\varphi_1(1, -1, 0, 2) = 0$ ,  $\varphi_1(0, 1, 1, -1) = 0$ ,  $\varphi_2(1, -1, 0, 2) = 1 + 1 - 2 = 0$ ,  $\varphi_2(0, 1, 1, -1) = -1 + 1 = 0$ . They are independent (compare the coefficients of  $x_3$  and  $x_4$ ). The spanning vectors of  $U$  are independent (compare first entries, then second), so  $\dim U^0 = 4 - 2 = 2$  by Theorem 4.2.2, and  $(\varphi_1, \varphi_2)$  is a basis by Theorem 1.6.4.

- (b) By Theorem 4.5.2,  $\psi_i(\mathbf{v} + U) = \varphi_i(\mathbf{v})$ . So  $\psi_1((2, 0, 1, 3) + U) = 2 + 0 - 1 = 1$  and  $\psi_2((2, 0, 1, 3) + U) = 2 - 0 - 3 = -1$ .
- (c) We need cosets  $\mathbf{c}_1, \mathbf{c}_2$  with  $\psi_i(\mathbf{c}_j) = \delta_{ij}$ . The vector  $-\mathbf{e}_3$  has  $\varphi_1(-\mathbf{e}_3) = 1$  and  $\varphi_2(-\mathbf{e}_3) = 0$ , and  $-\mathbf{e}_4$  has  $\varphi_1(-\mathbf{e}_4) = 0$ ,  $\varphi_2(-\mathbf{e}_4) = 1$ . Put  $\mathbf{c}_1 = -\mathbf{e}_3 + U$  and  $\mathbf{c}_2 = -\mathbf{e}_4 + U$ . If  $a\mathbf{c}_1 + b\mathbf{c}_2 = \mathbf{0}$ , applying  $\psi_1$  and  $\psi_2$  gives  $a = 0$  and  $b = 0$ ; so  $(\mathbf{c}_1, \mathbf{c}_2)$  is independent, and since  $\dim(\mathbb{R}^4/U) = 2$  it is a basis (Theorem 1.6.4). By construction  $\psi_i(\mathbf{c}_j) = \delta_{ij}$ , so  $(\psi_1, \psi_2)$  is its dual basis (Theorem 4.1.3).

### Exercise. B2: Functionals with the same kernel

Let  $V$  be a vector space over  $F$ , of any dimension, and let  $\varphi, \psi \in V^*$  be non-zero. Prove that  $\ker \varphi = \ker \psi$  if and only if  $\psi = c\varphi$  for some non-zero  $c \in F$ . (Exercise (C1: Functionals with the same kernel) proved  $(\Rightarrow)$  by a different route; here use the hyperplane decomposition.)

*Hint: the remark after Theorem 4.5.3.*

*Solution.*  $(\Leftarrow)$  If  $\psi = c\varphi$  with  $c \neq 0$ , then  $\psi(\mathbf{v}) = 0$  if and only if  $c\varphi(\mathbf{v}) = 0$ , if and only if  $\varphi(\mathbf{v}) = 0$ , since  $c$  is invertible. So  $\ker \psi = \ker \varphi$ .

$(\Rightarrow)$  Suppose  $\ker \varphi = \ker \psi = H$ . Since  $\varphi \neq 0$ , choose  $\mathbf{v}_0$  with  $\varphi(\mathbf{v}_0) \neq 0$ ; then  $\mathbf{v}_0 \notin H$ , so also  $\psi(\mathbf{v}_0) \neq 0$ . Put  $c = \psi(\mathbf{v}_0)/\varphi(\mathbf{v}_0)$ , which is non-zero, and  $\chi = \psi - c\varphi \in V^*$ . Then  $\chi$  vanishes on  $H$ , because both  $\psi$  and  $\varphi$  do, and  $\chi(\mathbf{v}_0) = \psi(\mathbf{v}_0) - c\varphi(\mathbf{v}_0) = 0$ . By Theorem 4.5.3  $H$  is a hyperplane, and by the remark after it  $V = H \oplus \text{span}(\mathbf{v}_0)$ . A linear map vanishing on  $H$  and on  $\mathbf{v}_0$  vanishes on every  $\mathbf{h} + a\mathbf{v}_0$ , hence on  $V$ . So  $\chi = 0$ , that is,  $\psi = c\varphi$ .

### Exercise. B3: Dimension computations

- (a) Let  $\dim V = 7$  and  $\dim U = 3$ . Find  $\dim U^*$ ,  $\dim(V^*/U^0)$ ,  $\dim(V/U)^*$  and  $\text{codim } U$ .
- (b) In  $V = F[x]_{\leq 3}$ , let  $U = \{p : p(0) = p(1) = 0\}$ . Using the map  $T: V \rightarrow F^2$ ,  $T(p) = (p(0), p(1))$ , find  $\text{codim } U$  and  $\dim U$ , and write  $U$  as an intersection of two hyperplanes.
- (c) Hence find a basis of  $U^0$  consisting of evaluation functionals.

*Solution.*

- (a)  $\dim U^* = 3$  by Corollary 4.1.4.  $\dim U^0 = 7 - 3 = 4$  by Theorem 4.2.2, so  $\dim(V^*/U^0) = 7 - 4 = 3$  (Theorem 3.9.5), consistent with Theorem 4.5.1.  $\dim(V/U)^* = \dim(V/U) = 4$ , consistent with Theorem 4.5.2, and  $\text{codim } U = 4$ .
- (b)  $T$  is linear because evaluation is (Theorem 0.8.5), and  $\ker T = U$ . It is surjective:  $T(1) = (1, 1)$  and  $T(x) = (0, 1)$  span  $F^2$ . By Theorem 3.9.7,  $V/U \cong F^2$ , so  $\text{codim } U = 2$ , and  $\dim U = 4 - 2 = 2$  by Theorem 3.9.5. Writing  $\varepsilon_0(p) = p(0)$  and  $\varepsilon_1(p) = p(1)$ , we have  $U = \ker \varepsilon_0 \cap \ker \varepsilon_1$ , and both kernels are hyperplanes by Theorem 4.5.3, since both functionals send 1 to 1.

- (c) Both  $\varepsilon_0$  and  $\varepsilon_1$  lie in  $U^0$ . They are independent: if  $a\varepsilon_0 + b\varepsilon_1 = 0$ , evaluating at  $x$  gives  $b = 0$ , and then evaluating at 1 gives  $a = 0$ . Since  $\dim U^0 = 4 - 2 = 2$ ,  $(\varepsilon_0, \varepsilon_1)$  is a basis of  $U^0$  (Theorem 1.6.4).

### 4.5.6.3 C. Going deeper

#### Exercise. C1: Subspaces of finite codimension are intersections of hyperplanes

Let  $V$  be a vector space over  $F$ , of any dimension, and let  $U$  be a subspace of finite codimension  $k \geq 1$ .

- (a) Prove that there are non-zero  $\varphi_1, \dots, \varphi_k \in U^0$  with  $U = \ker \varphi_1 \cap \dots \cap \ker \varphi_k$ . Deduce that  $U$  is an intersection of  $k$  hyperplanes.
- (b) Prove that  $U$  is not an intersection of fewer than  $k$  hyperplanes.

*Hint: for (a), take a dual basis of  $(V/U)^*$ ; for (b), consider the map  $V \rightarrow F^m$ ,  $\mathbf{v} \mapsto (\varphi_1(\mathbf{v}), \dots, \varphi_m(\mathbf{v}))$ .*

*Solution.*

- (a) Let  $\pi: V \rightarrow V/U$  be the quotient map. Since  $\dim(V/U) = k$ , choose a basis  $(\mathbf{c}_1, \dots, \mathbf{c}_k)$  of  $V/U$  and let  $(\psi_1, \dots, \psi_k)$  be its dual basis (Theorem 4.1.3). Put  $\varphi_i = \psi_i \circ \pi$ . By Theorem 4.5.2,  $\varphi_i \in U^0$ , and  $\varphi_i \neq 0$  because  $\pi'$  is injective and  $\psi_i \neq 0$ . Since each  $\varphi_i$  lies in  $U^0$ ,  $U \subseteq \ker \varphi_i$  for each  $i$ . Conversely, let  $\mathbf{v} \in \ker \varphi_1 \cap \dots \cap \ker \varphi_k$ , and write  $\mathbf{v} + U = \sum_j a_j \mathbf{c}_j$ . Applying  $\psi_i$  gives  $a_i = \psi_i(\mathbf{v} + U) = \varphi_i(\mathbf{v}) = 0$  for every  $i$ . So  $\mathbf{v} + U$  is the zero coset, and  $\mathbf{v} \in U$  by Lemma 3.9.3 (b). This proves  $U = \bigcap_i \ker \varphi_i$ . Each  $\ker \varphi_i$  is a hyperplane by Theorem 4.5.3.
- (b) Suppose  $U = H_1 \cap \dots \cap H_m$  with hyperplanes  $H_i$ . By Theorem 4.5.3,  $H_i = \ker \varphi_i$  for non-zero  $\varphi_i \in V^*$ . The map  $S: V \rightarrow F^m$ ,  $S(\mathbf{v}) = (\varphi_1(\mathbf{v}), \dots, \varphi_m(\mathbf{v}))$ , is linear, since each entry is, and  $\ker S = \bigcap_i \ker \varphi_i = U$ . By Theorem 3.9.7,  $V/U \cong \text{im } S$ , a subspace of  $F^m$ . Hence  $k = \dim(V/U) = \dim \text{im } S \leq m$  by Theorem 1.6.7. So at least  $k$  hyperplanes are needed.

#### Exercise. C2: Complements and a choice of copy of $U^*$

Let  $V$  be finite-dimensional, and let  $U, W$  be subspaces with  $V = U \oplus W$ .

- (a) Prove that  $V^* = U^0 \oplus W^0$  without using Theorem 4.2.3 (d). (This is Exercise (C3: Dual of a direct sum decomposition) (a), which used it.)
- (b) Deduce that the restriction map  $\rho: V^* \rightarrow U^*$  restricts to an isomorphism  $W^0 \rightarrow U^*$ .
- (c) In  $\mathbb{R}^2$ , let  $U = \text{span}(\mathbf{e}_1)$ ,  $W_1 = \text{span}(\mathbf{e}_2)$  and  $W_2 = \text{span}((1, 1))$ . Find  $W_1^0$  and  $W_2^0$ , and find the element of each that restricts to the functional  $\xi \in U^*$  with  $\xi(\mathbf{e}_1) = 1$ . Conclude that the copy of  $U^*$  inside  $V^*$  depends on the complement.

*Solution.*

- (a) Let  $n = \dim V$ . If  $\varphi \in U^0 \cap W^0$ , then  $\varphi$  vanishes on  $U$  and on  $W$ , hence on every  $\mathbf{u} + \mathbf{w}$ , that is, on  $U + W = V$ ; so  $\varphi = 0$ . By Theorem 4.2.2,  $\dim U^0 + \dim W^0 =$

$(n - \dim U) + (n - \dim W) = 2n - n = n$ , using  $\dim U + \dim W = n$  for a direct sum (Theorem 1.7.4). By Theorem 1.7.3,  $\dim(U^0 + W^0) = n - 0 = n = \dim V^*$ , so  $U^0 + W^0 = V^*$  (Theorem 1.6.8), and the sum is direct because the intersection is  $\{0\}$ .

- (b) The restriction of  $\rho$  to  $W^0$  is linear. Its kernel is  $W^0 \cap \ker \rho = W^0 \cap U^0 = \{0\}$  by Theorem 4.5.1 and (a), so it is injective. Its domain has dimension  $n - \dim W = \dim U = \dim U^*$ . An injective linear map between spaces of the same finite dimension is an isomorphism (Corollary 3.3.2 (e)).
- (c) Write functionals on  $\mathbb{R}^2$  as  $ax + by$ .  $W_1^0$  consists of those with  $b = 0$ , so  $W_1^0 = \text{span}(x)$ .  $W_2^0$  consists of those with  $a + b = 0$ , so  $W_2^0 = \text{span}(x - y)$ . Restricting to  $U$ :  $x|_U$  and  $(x - y)|_U$  both take the value 1 at  $\mathbf{e}_1$ , so both equal  $\xi$ . The elements  $x$  and  $x - y$  are different, so the subspaces  $W_1^0 \neq W_2^0$  of  $V^*$  are two different copies of  $U^*$ , one for each complement.